Republic of Rwanda
Ministry of Education

RTB | RWANDA TVET BOARD

ETEAI501

ARTIFICIAL INTELLIGENCE INTEGRATION INTO ELECTRONIC DEVICES

Competence: Integrate Artificial Intelligence into electronic devices

RQF Level: 5

Learning Hours
90 hours

Credits: 9

Sector: Technical service

Trade: Electronics and Telecommunication

Module Type: Specific

Curriculum: TSVETE5001-TVET Certificate V in Electronics and Telecommunication

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<table>
<tr><td>Purpose statement</td><td colspan="6">This module describes the knowledge, skills and attitude required to integrate artificial intelligence into electronic devices
It is intended to be pursued by learner at TVET certificate level V in Electronics and Telecommunication. At the end of this module, the learner will be able to Perform pre-AI training model activities, Train IA model and Deploy AI.</td></tr>
<tr><td>Learning assumed to be in place</td><td colspan="6">▪ Produce electronic smart devices circuit board
▪ Perform chip programming
▪ Produce electronic smart devices enclosures
▪ Apply python for machine learning</td></tr>
<tr><td rowspan="6">Delivery modality</td><td colspan="2">Training delivery</td><td>100%</td><td colspan="2">Assessment</td><td>Total 100%</td></tr>
<tr><td colspan="2">Theoretical content</td><td>30%</td><td rowspan="4">Formative assessment</td><td>30%</td><td rowspan="4">50%</td></tr>
<tr><td colspan="2">Practical work:</td><td rowspan="3">70%</td><td rowspan="3">70%</td></tr>
<tr><td>• Group work and presentation</td><td>30%</td></tr>
<tr><td>• Individual work</td><td>40%</td></tr>
<tr><td colspan="3"></td><td colspan="2">Summative Assessment</td><td>50%</td></tr>
</table>

Knowledge	Skills	Attitude
✓ Understanding of AI Technologies ✓ Hardware Compatibility ✓ Data Collection and Preprocessing ✓ Model Development and Training ✓ Integration with Device Software ✓ Real-time Inference and Optimization ✓ User Experience Design ✓ Security and Privacy Considerations ✓ Basic of electronic fundamentals ✓ Digital electronics	✓ AI and Machine Learning (ML) Fundamentals ✓ Programming Languages ✓ Embedded Systems and IoT ✓ Hardware Integration ✓ Data Processing and Management ✓ Real-time Inference Optimization ✓ Testing and Validation ✓ Computer skills	✓ Innovative ✓ Honest ✓ Accountability ✓ Self-motivated ✓ Time management ✓ Humble ✓ Creative ✓ Responsible ✓ Flexible ✓ integrity ✓ Goal oriented ✓ Self-confident ✓ Motivated ✓ Energetic ✓ Able to work independently ✓ Strong moral character ✓ Personal hygiene/grooming ✓ Time management ✓ Open-minded ✓ Organized ✓ Maintain health ✓ Positive work ethics ✓ Teamwork and Collaboration ✓

Elements of Competency and Performance Criteria

Elements of competence	Performance criteria
1. Perform pre-AI training model activities	1.1. AI project is appropriately documented in reference to the problem to be solved and AI documentation process
	1.2. AI project related Data are properly collected based on data categories, AI-Data collection techniques and Data pre-processing operations
	1.3. Data file is accurately created according to the Data formats, and file creation process
	1.4. AI dataset is effectively imported based on appropriate development tools, and types of dataset
2. Train IA model	2.1 Training Dataset is appropriately separated from Testing data according to the AI data splitting standard, accuracy parameters
	2.2 AI model is adequately trained according to Machine leaning model training procedures and algorithms
	2.3 AI Model performance is properly evaluated according to AI model test strategies and model accuracy improving techniques
	2.4 Trained model file is correctly exported as per model deployment file formats and AI file libraries
3. Deploy AI	3.1 Appropriate embedded hardware is properly selected according to embedded system board features and specifications
	3.2 Trained Model is accurately Optimized refer to embedded platform specifications and model optimization techniques
	3.3 Optimized model file is properly imported into existing program source code based on used Integrated Development Environment (IDE), AI model integration tools and AI model integration process

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Perform pre-AI training model activities 2. Train IA model 3. Deploy AI.
4. **Learning outcome 1: Perform pre-AI training model activities**	**Learning hours: 25 hours**

Indicative content

- **Description of AI concepts**
  - ✓ Definitions of key terms
  - ✓ Historical
  - ✓ Importance
- **Documentation of AI project**
  - ✓ **AI project source of information**
  - ✓ **AI vs Problems to be solved**
    - Social
    - Economical
    - technical
  - ✓ AI documentation process
- **Collection of AI project related Data**
  - ✓ AI data categories,
  - ✓ AI-Data collection techniques
  - ✓ Data pre-processing(cleaning, filtering, and Data transformation)
- **Creation of AI Data file**
  - ✓ Data formats
  - ✓ File creation process
- **Import of AI dataset**

- ✓ Development tools
  - Data import software
    - Environment software (anaconda, Visual Studio Code)
  - DATASET Importing techniques
- ✓ Types of dataset

Resources required for the learning outcome	
Equipment	Computer , projectors, camera, mobile phone
Materials	Papers, marker pens, flips chart, pens, chalks,
Tools	Software (anaconda navigator , spyder , VS Code,….)
Facilitation techniques	▪ Brainstorming ▪ Research ▪ presentation ▪ Demonstration ▪ Practical exercise ▪ Trainer guided ▪ Group discussion
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment

Learning outcome 2: Train IA model	Learning hours: 35 hours
Indicative content	
• **Data splitting** ✓ Training Dataset ✓ Testing data ✓ Dataset Splitting standards Splitting ratio	

- Overfitting
- Underfitting
- Unbiased Evaluation
- Data Variability

✓ Accuracy parameters

- Accuracy
- Precision
- Recall
- F1 score
- Confusion Matrix
- Root mean square error (RMSE)
- Mean absolute Error (MAE)

- **Train AI model/ machine learning model**
  - ✓ Description of Machine learning model
    - Definition of AI model
    - Importance of Model Training
    - Types of machine learning
  - ✓ AI model training procedures
  - ✓ AI model training algorithms
    - Selection of algorithm to train a model
- **Evaluate AI Model performance**
  - ✓ **AI model testing**
    - Types of ML Testing
    - Evaluation Metrics for ML Models
    - Ethical Considerations in ML Testing
  - ✓ **Model accuracy improving techniques**
    - Data Quality Improvement
    - Data Augmentation
    - Feature Engineering
    - Model Selection/ Tuning or Try different algorithm
    - Hyperparameter Tuning

• **Export Trained model as a file** ✓ File formats ✓ AI file libraries

Resources required for the learning outcome	
Equipment	Computer, projectors, camera, mobile phone
Materials	Papers, marker pens, flips chart, pens, chalks
Tools	Software (anaconda navigator, spyder, VS Code….)
Facilitation techniques	▪ Brainstorming ▪ Research ▪ presentation ▪ Demonstration ▪ Practical exercise ▪ Trainer guided ▪ Group discussion
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment

Learning outcome 3: Deploy AI	Learning hours: 30 hours
Indicative content	

- **Selection of embedded hardware**
  - ✓ Hardware specification and features.
    - Processor
    - Memory
    - Power requirement
    - On-board accelerators (NPU/GPUs)
    - I/0 Interfaces
    - OS/RTOS supported by hardware for ML framework compatibility
    - Supported embedded hardware pricing
  - ✓ Microcontroller and single board computers for ML
    - Arduino Nano 33 BLE Sense
    - ESP-Eye
    - ESP32-CAM
    - Raspberry Pi Boards
    - Nvidia Jetson Nano

- **Optimize Trained Model to embedded platform**
  - ✓ The reason behind model optimization
    - Memory constraints
    - Power limitation
    - Small Computational resources at the edge
  - ✓ Model optimization techniques
    - Quantization
    - Pruning
    - Clustering

- **Deploy Optimized model file is properly**
  - ✓ Imported AI model into existing program source code
    - Compile model to C/C++ code or python
    - Import a compiled model into your source code
  - ✓ AI model integration tools
    - Integrated Development Environment (IDE)
    - Edge Impulse
    - PyTorch
    - TensorFlow
  - ✓ AI model integration process

Resources required for the learning outcome	
Equipment	Computer, projectors, camera, mobile phone, Microcontrollers,
Materials	Papers, marker pens, flips chart, pens, chalks
Tools	Software (anaconda navigator, spyder, VS Code, Pytorch, Edge impulse)
Facilitation techniques	▪ Brainstorming ▪ Research ▪ presentation ▪ Demonstration ▪ Practical exercise ▪ Trainer guided ▪ Group discussion
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment
Integrated/Summative assessment	

Integrated situation

XYZ Company, located in Kigali city, has faced several security breaches in their store rooms. The challenge is to implement a Smart Surveillance System that can intelligently differentiate between thieves (people) and pets (dogs, cats, rats). The system should activate alarms only when a human intruder is detected, thus avoiding false alarms caused by pets.

Areas to be Monitored:

- Store room entrances
- Storage racks

Task

Smart Surveillance System to Differentiate Thieves from Pets

Resources

Tools	Software (anaconda navigator, spyder, VS Code, Pytorch, Edge impulse)
Equipment	• Computer with Arduino IDE and Python environment • ESP32-CAM modules • Breadboards and jumper wires • MicroSD cards • USB cables • Power supply units • Router for Wi-Fi connection
Materials/ Consumables	Papers, marker pens, flips chart, pens, chalks

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
	1.1. AI project is appropriately documented in reference to the problem to be	AI project is well documented			5
		AI documentation processes are respected			5

1. Perform Pre-AI training model activities (40%)	solved and AI documentation process				
	1.2. AI project related Data are properly collected based on data categories, AI-Data collection techniques and Data pre-processing operations	Data collection techniques are respected			5
		Data are well collected			5
		Data preprocessing stage are well performed ❖ Cleaning ❖ Filtering ❖ Transformation			10
	1.3. Data file is accurately created according to the Data formats, and file creation process	Data formats are respected during creation of a file			5
	1.4. AI dataset is effectively imported based on appropriate development tools, and types of datasets	Importation flow of dataset is respected.			5
2. **Train AI model (50%)**	2.1. Training Dataset is appropriately separated from Testing data according to the AI data splitting standard, accuracy parameters	Splitting ratio are followed based to training and test dataset			4
		Overfitting is checked and resolved properly			2
		Underfitting is checked and resolved properly			2
		Bias is evaluated			2
	2.2 AI model is adequately trained according to Machine leaning model training procedures and algorithms	Model training procedures are followed			5
		ML algorithm is selected according to the problem to be solved.			5
		Accuracy is checked			5

	2.3. AI Model performance is properly evaluated according to AI model test strategies and model accuracy improving techniques	Precision is checked			2
		Recall is checked			2
		F1 score is checked			2
		Confusion Matrix is checked			2
		Model improvement is performed			5
	2.4. Trained model file is correctly exported as per model deployment file formats and AI file libraries	Libraries have been well selected to export the model			5
		File format has been selected according to the device to use for deployment.			7
3. Deploy AI (10%)	3.1. Appropriate embedded hardware is properly selected according to embedded system board features and specifications	Hardware is selected according to the problem to be solved			2
	3.2. Trained Model is accurately Optimized refer to embedded platform specifications and model optimization techniques	Model optimization techniques are applied.			2
	3.3. Optimized model file is properly imported into existing program source code based on used Integrated Development Environment (IDE), AI model integration tools and AI model integration process	AI model is compiled to C/C++			2
		AI model integration tools are selected			2
		AI model is correctly integrated to the electronic device.			2
Total marks					**100**
Percentage Weightage					100%
Minimum Passing line % (Aggregate):					70%

List of abbreviations

AI - Artificial Intelligence

RQF - Regulated Qualifications Framework

TVET - Technical and Vocational Education and Training

IDE - Integrated Development Environment

NPU - Neural Processing Unit

GPU - Graphics Processing Unit

OS - Operating System

RTOS - Real-Time Operating System

ML - Machine Learning

RMSE - Root Mean Square Error

MAE - Mean Absolute Error

CAT - Continuous Assessment Test

ESP32 - A series of low-cost, low-power system on a chip microcontroller with integrated Wi-Fi and dual-mode Bluetooth

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Glossary:

Activity: Activities include releases, events, and deployment plans that you develop, start, and complete with the product.

Application environment: A user-defined collection of resources that hosts an application. These application environments refer to environments that are created to be in the product.

Application process: A process that is associated with an application. Unlike a component or generic process, an application process is created from application-level steps.

Application: One or more computer programs or software components that provide a function in direct support of a specific business process or processes. See also application server.

Architecture: The internal structure of a computer system or a chip that determines its operational functionality and performance.

Control processor: The processor in a processor array machine that issues the instructions to be executed by all the processors in the processor array. Alternatively, the control processor may perform tasks in which the processors in the array are not involved, e.g., I/O operations or serial operations.

Deployment: The activities used to deliver a software project to a deployment target. Typically, you run deployments for each stage of your release lifecycle, ending with the production stage when the software becomes generally available.

Duration: The time a task takes to run. Duration is measured from the time a task starts until it is resolved. When you create some task types, you can estimate their expected duration. Duration is reported in minutes.

Environment: A collection of resources that identify the components that can be deployed by the parent application and the agents that do the work.

Events: Release-related activities that are applied to releases and tracked with a calendar. You can use events to organize your releases and other time-dependent activities, such as holidays and blackouts.

Initiative: An action to take for the change occurred.

Insights: A set of related features such as metric data dashboards, reports, and the application portfolio.

Lifecycle: The phases in a release. A lifecycle is a template for the stages of work in a release.

Plugin: A separately installable software module that adds functionality to an existing program, application, or interface.

Resource: A user-defined construct that is based on the architectural model of IBM Urban Code Velocity. A resource represents a deployment target.

Role: A job function that identifies the tasks that a user can perform and the resources to which a user has access. A user can be assigned one or more roles. Processors have direct access to all the memory in the system. Because of technological limitations on shared bandwidth generally, not more than about 16 processors share a common memory.

Task: Represents a business-meaningful activity that has the starting and ending points and a measurable duration. Durations are used to estimate deployment times. You add tasks to deployment plans. When you run a deployment, you complete the tasks in the plan

User: A representation of an account on the server. Users can be members of teams and groups. Users can be created in IBM Urban Code Velocity server or import users from an external authentication realm, including an LDAP, Active Directory, or SSO provider